

02/05/2024



Aakash

Medical | IIT-JEE | Foundations

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Phase-I
CODE-A

FINAL TEST SERIES for NEET-2024

MM : 720

Test -13

Time : 3 Hrs. 20 Mins.

Complete Syllabus of Class XI & XII

Instructions :

- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION - A

1. Match List-I with List-II.

	List-I (Electromagnetic waves)		List-II (Wavelength)
(a)	AM radio waves	(i)	10^{-10} m
(b)	Microwaves	(ii)	10^2 m
(c)	Infrared radiations	(iii)	10^{-2} m
(d)	X-rays	(iv)	10^{-4} m

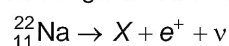
Choose the correct answer from the options given below.

- (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
 - (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
 - (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
 - (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
- An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of 1.5 ms^{-1} . The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is ($g = 10 \text{ m s}^{-2}$)

(1) 34500 (2) 23500

(3) 23000 (4) 20000

3. In the given nuclear reaction, the element X is

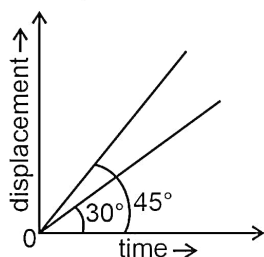
(1) ${}_{10}^{22}\text{Ne}$ (2) ${}_{12}^{22}\text{Mg}$ (3) ${}_{11}^{23}\text{Na}$ (4) ${}_{10}^{23}\text{Ne}$ 4. Two hollow conducting spheres of radii R_1 and R_2 ($R_1 \gg R_2$) have equal positive charges. The potential would be

- Equal on both the spheres
- Dependent on the material property of the sphere
- More on bigger sphere
- More on smaller sphere

5. If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is

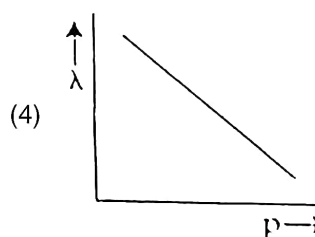
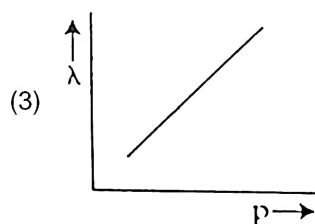
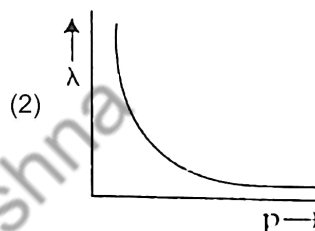
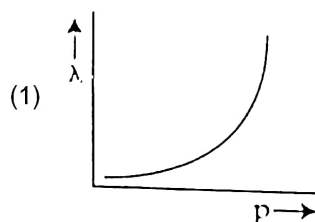
- $1 : \sqrt{2}$ (2) $1 : 2$
- $1 : 1$ (4) $\sqrt{2} : 1$

6. If a soap bubble expands, the pressure inside the bubble
- Remains the same
 - Is equal to the atmospheric pressure
 - Decreases
 - Increases
7. The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second.
- 1 : 3 : 5 : 7
 - 1 : 1 : 1 : 1
 - 1 : 2 : 3 : 4
 - 1 : 4 : 9 : 16
8. A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is
- 12.56×10^{-4} T
 - 6.28×10^{-4} T
 - 6.28×10^{-2} T
 - 12.56×10^{-2} T
9. The displacement-time graphs of two moving particles make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is



- 1 : 2
 - $1 : \sqrt{3}$
 - $\sqrt{3} : 1$
 - 1 : 1
10. The peak voltage of the ac source is equal to
- $\sqrt{2}$ times the rms value of the ac source
 - $1/\sqrt{2}$ times the rms value of the ac source
 - The value of voltage supplied to the circuit
 - The rms value of the ac source

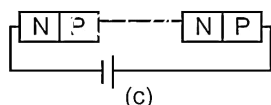
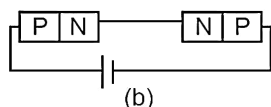
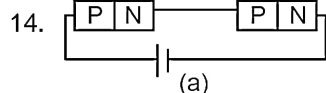
11. Let T_1 and T_2 be the energy of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is
- 4 : 9
 - 9 : 4
 - 1 : 4
 - 4 : 1
12. The graph which shows the variation of the de Broglie wavelength (λ) of a particle and its associated momentum (p) is



Space for Rough Work

13. Plane angle and solid angle have

- (1) No units and no dimensions
- (2) Both units and dimensions
- (3) Units but no dimensions
- (4) Dimensions but no units



In the given circuits (a), (b) and (c), the potential drop across the two p - n junctions are equal in

- (1) Circuit (c) only
- (2) Both circuits (a) and (c)
- (3) Circuit (a) only
- (4) Circuit (b) only

15. The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is

- (1) 4 : 1
- (2) 1 : $\sqrt{2}$
- (3) 2 : 1
- (4) $\sqrt{2}$: 1

16. The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s^2 is

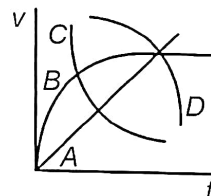
- (1) 12π
- (2) 104π
- (3) 2π
- (4) 4π

17. A copper wire of length 10 m and radius $\left(\frac{10^{-2}}{\sqrt{\pi}}\right)$ m has electrical resistance of 10 Ω . The

current density in the wire for an electric field strength of 10 (V/m) is

- (1) 10^{-5} A/m^2
- (2) 10^5 A/m^2
- (3) 10^4 A/m^2
- (4) 10^6 A/m^2

18. A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball (v) as a function of time (t) is



- (1) C
- (2) D
- (3) A
- (4) B

19. Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is

- (1) 1 : 4
- (2) 4 : 1
- (3) 1 : 2
- (4) 2 : 1

20. The maximum kinetic energy of photoelectrons emitted from a metal surface, when a light of wavelength λ is incident on it, is 1 eV. When a light of wavelength $\frac{\lambda}{3}$ is incident on the surface,

maximum kinetic energy becomes 4 times. The work function of metal is

- (1) $\frac{hc}{\lambda}$
- (2) $\frac{hc}{3\lambda}$
- (3) $\frac{2hc}{\lambda}$
- (4) $\frac{hc}{2\lambda}$

21. In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is

- (1) 9
- (2) 12
- (3) 6
- (4) 8

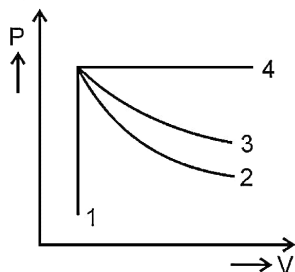
Space for Rough Work

22. When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r , the velocity of light, v is given by (c -velocity of light in vacuum)

$$(1) v = \sqrt{\frac{\epsilon_r}{\mu_r}} \quad (2) v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$$

$$(3) v = c \quad (4) v = \sqrt{\frac{\mu_r}{\epsilon_r}}$$

23. An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is



- (1) 3 (2) 4
(3) 1 (4) 2
24. As the temperature increases, the electrical resistance
- (1) Increases for conductors but decreases for semiconductors
 - (2) Decreases for conductors but increases for semiconductors
 - (3) Increases for both conductors and semiconductors
 - (4) Decreases for both conductors and semiconductors
25. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is

- (1) 10 m (2) 5 m
(3) $\frac{10}{3}$ m (4) $\frac{20}{3}$ m

26. In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be.

- (1) 60 Hz (2) 120 Hz
(3) Zero (4) 30 Hz

27. Given below are two statements:

Statement I : Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current carrying conductor only.

Statement II : Biot-Savart's law is analogous to Coulomb's inverse square law of charge q , with the former being related to the field produced by a scalar source, Idl while the latter being produced by a vector source, q .

In light of above statements choose the most appropriate answer from the options given below

- (1) Statement I is correct and Statement II is incorrect
(2) Statement I is incorrect and Statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

28. The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is

- (1) 36×10^5 J (2) 1×10^5 J
(3) 36×10^7 J (4) 36×10^4 J

29. A body of mass 60 g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is

- (1) 20 N/kg (2) 180 N/kg
(3) 0.05 N/kg (4) 50 N/kg

Space for Rough Work

30. A biconvex lens has radii of curvature, 20 cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is

- (1) +5 D (2) Infinity
(3) +2 D (4) +20 D

31. A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v , the speed of the third (lighter) fragment is

- (1) $2\sqrt{2}v$ (2) $3\sqrt{2}v$
(3) v (4) $\sqrt{2}v$

32. The dimensions $[MLT^{-2}A^{-2}]$ belong to the

- (1) Magnetic permeability
(2) Electric permittivity
(3) Magnetic flux
(4) Self inductance

33. The angle between the electric lines of force and the equipotential surface is

- (1) 90° (2) 180°
(3) 0° (4) 45°

34. A square loop of side 1 m and resistance $1\ \Omega$ is placed in a magnetic field of 0.5 T. If the plane of loop is perpendicular to the direction of magnetic field, the magnetic flux through the loop is

- (1) 1 weber (2) Zero weber
(3) 2 weber (4) 0.5 weber

35. A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60° . The angle between the refracted and reflected rays would be

- (1) 90° (2) 120°
(3) 30° (4) 60°

SECTION-B

36. A ball is projected with a velocity, 10 ms^{-1} , at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be

- (1) 5 ms^{-1}
(2) 10 ms^{-1}
(3) Zero
(4) $5\sqrt{3}\text{ ms}^{-1}$

37. A series LCR circuit with inductance 10 H, capacitance $10\ \mu\text{F}$, resistance $50\ \Omega$ is connected to an ac source of voltage, $V = 200\sin(100t)$ volt. If the resonant frequency of the LCR circuit is ν_0 and the frequency of the ac source is ν , then

- (1) $\nu_0 = \frac{50}{\pi}\text{ Hz}$, $\nu = 50\text{ Hz}$
(2) $\nu = 100\text{ Hz}$; $\nu_0 = \frac{100}{\pi}\text{ Hz}$
(3) $\nu_0 = \nu = 50\text{ Hz}$
(4) $\nu_0 = \nu = \frac{50}{\pi}\text{ Hz}$

38. The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is

- (1) 1382.5 (2) 14×10^2
(3) 138×10^1 (4) 1382

39. Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is

- (1) 10 (2) 8
(3) 11 (4) 9

Space for Rough Work

40. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

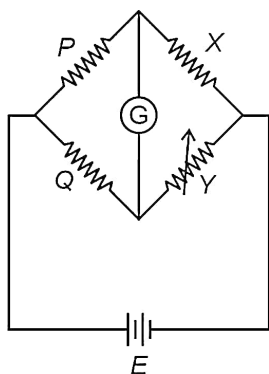
Assertion (A): The stretching of a spring is determined by the shear modulus of the material of the spring.

Reason (R): A coil spring of copper has more tensile strength than a steel spring of same dimensions.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

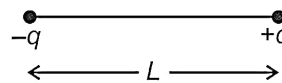
- (1) (A) is true but (R) is false
- (2) (A) is false but (R) is true
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (4) Both (A) and (R) are true and (R) is not the correct explanation of (A)

41. A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X , the resistances P and Q



- (1) Should be very large and unequal
- (2) Do not play any significant role
- (3) Should be approximately equal to $2X$
- (4) Should be approximately equal and are small

42. Two-point charges $-q$ and $+q$ are placed at a distance of L , as shown in the figure.



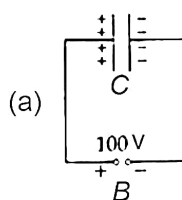
The magnitude of electric field intensity at a distance R ($R \gg L$) varies as

- (1) $\frac{1}{R^4}$
- (2) $\frac{1}{R^6}$
- (3) $\frac{1}{R^2}$
- (4) $\frac{1}{R^3}$

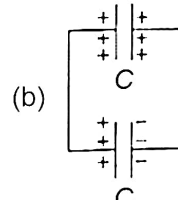
43. A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is

- (1) 5 : 4
- (2) 25 : 16
- (3) 1 : 1
- (4) 4 : 5

44. A capacitor of capacitance $C = 900 \text{ pF}$ is charged fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance $C = 900 \text{ pF}$ as shown in figure (b). The electrostatic energy stored by the system (b) is



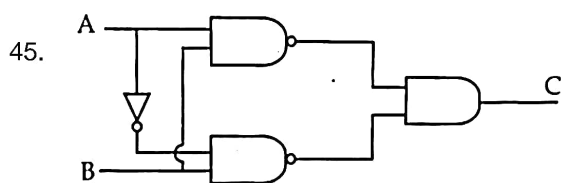
(a)



(b)

- (1) $2.25 \times 10^{-6} \text{ J}$
- (2) $1.5 \times 10^{-6} \text{ J}$
- (3) $4.5 \times 10^{-6} \text{ J}$
- (4) $3.25 \times 10^{-6} \text{ J}$

Space for Rough Work



The truth table for the given logic circuit is

(1)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

(2)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

(3)

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

(4)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

46. The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is

- (1) $5.6 \times 10^{-3} \text{ m}^3$ (2) 5.6 m^3
 (3) $5.6 \times 10^6 \text{ m}^3$ (4) $5.6 \times 10^3 \text{ m}^3$

47. Match List-I with List-II.

	List-I		List-II
(a)	Gravitational constant (G)	(i)	$[L^2T^{-2}]$
(b)	Gravitational potential energy	(ii)	$[M^{-1}L^3T^{-2}]$
(c)	Gravitational potential	(iii)	$[LT^{-2}]$
(d)	Gravitational intensity	(iv)	$[ML^2T^{-2}]$

Choose the correct answer from the options given below.

- (1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)
 (2) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)
 (3) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
 (4) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

48. From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is

- (1) A linearly increasing function of distance r upto the boundary of the wire and then decreasing one with $\frac{1}{r}$ dependence for the outside region
 (2) A linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.
 (3) Uniform and remains constant for both the regions
 (4) A linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region

49. A big circular coil of 1000 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1} . If the vertical component of earth's magnetic field at that place is $2 \times 10^{-5} \text{ T}$ and electrical resistance of the coil is 12.56Ω , then the maximum induced current in the coil will be

- (1) 1 A (2) 2 A
 (3) 0.25 A (4) 1.5 A

50. Two transparent media A and B are separated by a plane boundary. The speed of light in those media are $1.5 \times 10^8 \text{ m/s}$ and $2.0 \times 10^8 \text{ m/s}$, respectively. The critical angle for a ray of light for these two media is

- (1) $\tan^{-1}(0.500)$
 (2) $\tan^{-1}(0.750)$
 (3) $\sin^{-1}(0.500)$
 (4) $\sin^{-1}(0.750)$

Space for Rough Work

CHEMISTRY

SECTION - A

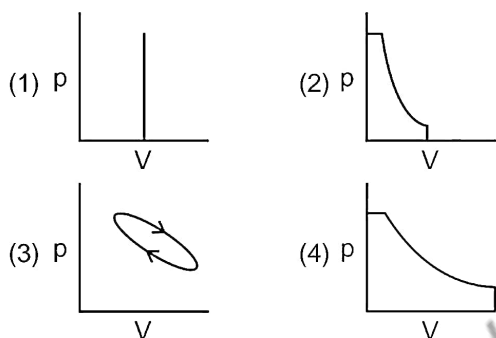
51. Choose the correct statement.

- (1) Carbon in diamond is sp^3 hybridised and in graphite is sp^2 hybridized
- (2) Both diamond and graphite are used as dry lubricants
- (3) Diamond and graphite have two dimensional network
- (4) Diamond is covalent and graphite is ionic

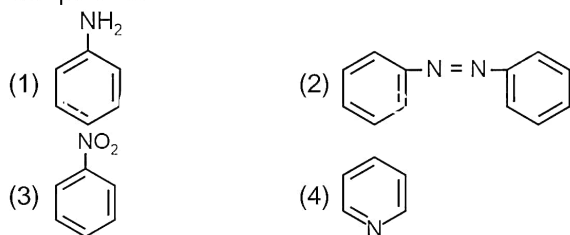
52. 2 moles of an ideal gas at 300 K is expanded isothermally from 1 L to 5 L volume. ΔU for this process is (use $R = 8.134 \text{ JK}^{-1}\text{mol}^{-1}$)

- (1) 4900 J
- (2) 0 J
- (3) 1284 J
- (4) 5040 J

53. Which of the following p-V curve represents maximum work done?



54. The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds?



55. K_H value for some gases at the same temperature 'T' are given:

Gas	K_H (kbar)
He	144.97
H ₂	69.16
N ₂	76.48
O ₂	34.86

Where K_H is Henry's Law constant in water. The order of their solubility in water is

- (1) $O_2 < H_2 < N_2 < He$
- (2) $He < N_2 < H_2 < O_2$
- (3) $He < H_2 < N_2 < O_2$
- (4) $O_2 < N_2 < H_2 < He$

56. Given below are two statements

Statement I: The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

Statement II: o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

In the light of the above statements, choose the most **appropriate** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

Space for Rough Work

57. Match List-I with List-II.

	List-I (Quantum Number)		List-II (Orbital)
(a)	$n = 2, l = 1$	(i)	3s
(b)	$n = 3, l = 2$	(ii)	2s
(c)	$n = 3, l = 0$	(iii)	2p
(d)	$n = 2, l = 0$	(iv)	3d

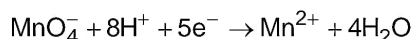
Choose the correct answer from the options given below.

- (1) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)
 (2) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)
 (3) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)
 (4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

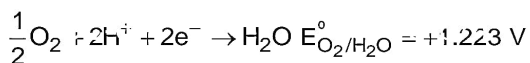
58. Which of the following set of molecules will have zero dipole moment?

- (1) Boron trifluoride, carbon tetrachloride, methane, 1,4-dichlorobenzene
 (2) Nitrogen trifluoride, ammonia, water, 1,3-dichlorobenzene
 (3) Boron trifluoride, carbon dioxide, 1,4-dichlorobenzene, ammonia
 (4) Water, carbon tetrachloride, ammonia, 1,3-dichlorobenzene

59. Given below are half cell reactions:



$$E_{\text{Mn}^{2+}/\text{MnO}_4^-}^\circ = -1.510 \text{ V}$$



Will the permanganate ion, MnO_4^- liberate O_2 from water in the presence of an acid?

(1) Yes, because $E_{\text{cell}}^\circ = +2.733 \text{ V}$ (2) No, because $E_{\text{cell}}^\circ = -2.733 \text{ V}$ (3) Yes, because $E_{\text{cell}}^\circ = +0.287 \text{ V}$ (4) No, because $E_{\text{cell}}^\circ = -0.287 \text{ V}$

60. Match List-I with List-II.

	List-I (Products formed)		List-I (Reaction of carbonyl compound with)
(a)	Cyanohydrin	(i)	NH_2OH
(b)	Acetal	(ii)	RNH_2
(c)	Schiff's	(iii)	Alcohol
(d)	Oxime	(iv)	HCN

Choose the correct answer from the options given below

- (1) (a) – (i), (b) – (iii), (c) – (ii), (d) – (iv)
 (2) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)
 (3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)
 (4) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)

61. The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is [Given pK_a of $\text{CH}_3\text{COOH} = 4.57$]

- (1) 4.57 (2) 2.57
 (3) 5.57 (4) 3.57

62. Dihedral angle of most stable conformer of n-butane is

- (1) 0° (2) 120°
 (3) 180° (4) 60°

63. The **incorrect** statement regarding enzymes is

- (1) Enzymes are polysaccharides
 (2) Enzymes are very specific for a particular reaction and substrate
 (3) Enzymes are biocatalysts
 (4) Like chemical catalysts enzymes reduce the activation energy of bio processes

Space for Rough Work

64. The major product formed in dehydrohalogenation reaction of 2-Bromobutane is But-1-ene. This product formation is based on

- (1) Huckel's Rule (2) Saytzeff's Rule
(3) Hund's Rule (4) Hoffmann Rule

65. Which of the following statement is not correct about diborane?

- (1) The four terminal hydrogen atoms and the two boron atoms lie in one plane
(2) Both the boron atoms are sp^2 hybridised
(3) There are two 3-centre-2-electron bonds
(4) The four terminal B-H bonds are two centre two electron bonds

66. Which one is not correct mathematical equation for Dalton's Law of partial pressure? Here p = total pressure of gaseous mixture

where p_i = partial pressure of i^{th} gas

χ_i = mole fraction of i^{th} gas in gaseous mixture

p_i° = pressure of i^{th} gas in pure state

- (1) $p_i = \chi_i p_i$
(2) $p_i = \chi_i p_i^\circ$
(3) $p = p_1 + p_2 + p_3$
(4) $p = n_1 \frac{RT}{V} + n_2 \frac{RT}{V} + n_3 \frac{RT}{V}$

67. The IUPAC name of the complex

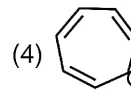
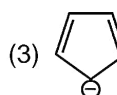
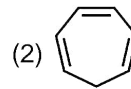
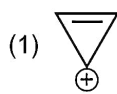
$[\text{Ag}(\text{H}_2\text{O})_2][\text{Ag}(\text{CN})_2]$ is

- (1) Dicyanidosilver(I) diaquaargentate(I)
(2) Diaquasilver(I) dicyanidoargentate(I)
(3) Dicyanidosilver(II) diaquaargentate(II)
(4) Diaquasilver(II) dicyanidoargentate(II)

68. Amongst the following which one will have maximum 'lone pair - lone pair' electron repulsions?

- (1) SF_4 (2) XeF_2
(3) ClF_3 (4) IF_5

69. Which compound amongst the following is **not** an aromatic compound?



70. Given below are two statements

Statement I : The boiling points of the following hydrides of group 16 elements increases in the order – $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$.

Statement II : The boiling points of these hydrides increase with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) **Statement I** is correct but **Statement II** is incorrect
(2) **Statement I** is incorrect but **Statement II** is correct
(3) Both **Statement I** and **Statement II** are correct
(4) Both **Statement I** and **Statement II** are incorrect

71. In one molal solution that contains 0.5 mole of a solute, there is

- (1) 100 mL of solvent
(2) 1000 g of solvent
(3) 500 mL of solvent
(4) 500 g of solvent

Space for Rough Work

72. Given below are two statements

Statement I: Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

Statement II: Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

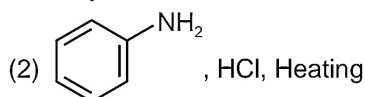
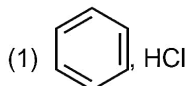
In the light of the above statements, choose the most **appropriate** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

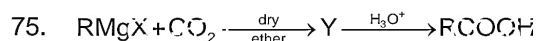
73. Gadolinium has a low value of third ionisation enthalpy because of

- (1) High electronegativity
- (2) High basic character
- (3) Small size
- (4) High exchange enthalpy

74. Which of the following sequence of reactions is suitable to synthesize chlorobenzene?



- (3) Benzene, Cl_2 , anhydrous FeCl_3
- (4) Phenol, NaNO_2 , HCl, CuCl



Y in the above reaction is

- (1) RCOO^-X^+
- (2) $(\text{RCOO})_2\text{Mg}$
- (3) $\text{RCOO}^-\text{Mg}^+\text{X}$
- (4) $\text{R}_3\text{CO}^-\text{Mg}^+\text{X}$

76. Identify the **incorrect** statement from the following.

- (1) In an atom, all the five 3d orbitals are equal in energy in free state
- (2) The shapes of d_{xy} , d_{yz} and d_{zx} orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other
- (3) All the five 5d orbitals are different in size when compared to the respective 4d orbitals
- (4) All the five 4d orbitals have shapes similar to the respective 3d orbitals

77. The IUPAC name of an element with atomic number 119 is

- (1) Unununnium
- (2) Ununoctium
- (3) Ununennium
- (4) Unnilennium

78. Given below are two statements :

Statement I : The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular masses because of stronger molecular association in aldehydes and ketones due to dipole - dipole interactions.

Statement II : The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular masses due to the absence of H-bonding.

In the light of the above statements, choose the **most appropriate** answer from the given below.

- (1) **Statement I** is correct but **Statement II** is incorrect
- (2) **Statement I** is incorrect but **Statement II** is correct
- (3) Both **Statement I** and **Statement II** are correct
- (4) Both **Statement I** and **Statement II** are incorrect

Space for Rough Work

79. What mass of 95% pure CaCO_3 will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction?



[Calculate upto second place of decimal point]

- (1) 3.65 g
- (2) 9.50 g
- (3) 1.25 g
- (4) 1.32 g

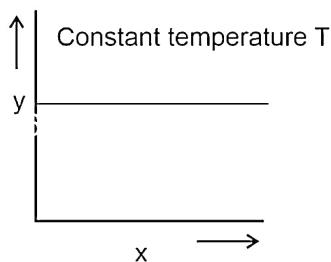
80. Match List-I with List-II.

	List-I (Reaction)		List-II (Product formed)
(a)	Gabriel synthesis	(i)	Ethers
(b)	Williamson synthesis	(ii)	Primary amines
(c)	Stephens reduction	(iii)	Salicylic acid
(d)	Kolbe reaction	(iv)	Aldehydes

Choose the correct answer from the options given below.

- (1) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)
- (2) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
- (3) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)
- (4) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

81. The given graph is a representation of kinetics of a reaction.



The y and x axes for zero and first order reactions, respectively are

- (1) Zero order ($y = \text{rate}$ and $x = \text{concentration}$), first order ($y = t_{1/2}$ and $x = \text{concentration}$)
- (2) Zero order ($y = \text{rate}$ and $x = \text{concentration}$), first order ($y = \text{rate}$ and $x = t_{1/2}$)
- (3) Zero order ($y = \text{concentration}$ and $x = \text{time}$), first order ($y = t_{1/2}$ and $x = \text{concentration}$)
- (4) Zero order ($y = \text{concentration}$ and $x = \text{time}$), first order ($y = \text{rate constant}$ and $x = \text{concentration}$)

82. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): ICl is more reactive than I_2 .

Reason (R): I-Cl bond is weaker than I-I bond.

In the light of the above statements, choose the most **appropriate** answer from the options given below:

- (1) (A) is correct but (R) is not correct
- (2) (A) is not correct but (R) is correct
- (3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

83. Which amongst the following is **incorrect** statement?

- (1) H_2^+ ion has one electron
- (2) O_2^+ ion is diamagnetic
- (3) The bond orders of O_2^+ , O_2 , O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1 respectively
- (4) C_2 molecule has four electrons in its two degenerate π molecular orbitals

Space for Rough Work

84. At 298 K, the standard electrode potentials of $\text{Cu}^{2+} / \text{Cu}$, $\text{Zn}^{2+} / \text{Zn}$, $\text{Fe}^{2+} / \text{Fe}$ and Ag^+ / Ag are 0.34 V, -0.76 V, -0.44 V and 0.80 V, respectively.

On the basis of standard electrode potential, predict which of the following reaction cannot occur?

- (1) $\text{FeSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Fe}(\text{s})$
 - (2) $2\text{CuSO}_4(\text{aq}) + 2\text{Ag}(\text{s}) \rightarrow 2\text{Cu}(\text{s}) + \text{Ag}_2\text{SO}_4(\text{aq})$
 - (3) $\text{CuSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$
 - (4) $\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$
85. The **incorrect** statement regarding chirality is
- (1) Enantiomers are superimposable mirror images on each other
 - (2) A racemic mixture shows zero optical rotation
 - (3) $\text{S}_{\text{N}}1$ reaction yields mixture of both enantiomers
 - (4) The product obtained by $\text{S}_{\text{N}}2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration

SECTION - B

86. Given below are two statements :

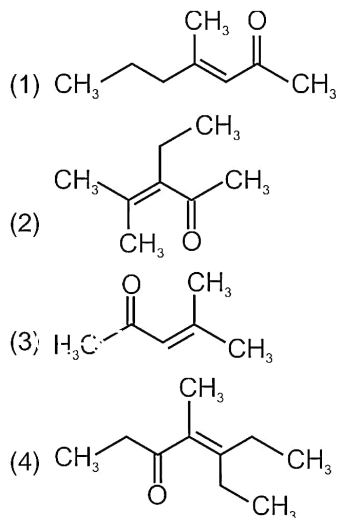
Statement I : The acidic strength of cresol is lower than phenol because of electron donating alkyl group.

Statement II : O-cresol, m-cresol and p-cresol have same acidic strength as they have one alkyl group attached to the phenolic ring.

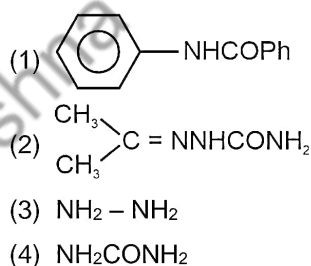
In the light of the above statements, choose the **most appropriate** answer from the given below.

- (1) **Statement I** is correct but **Statement II** is incorrect
- (2) **Statement I** is incorrect but **Statement II** is correct
- (3) Both **Statement I** and **Statement II** are correct
- (4) Both **Statement I** and **Statement II** are incorrect

87. Which one of the following is not formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating?



88. Lassaigne's test for the detection of nitrogen fail in the case of



89. Compound X on reaction with O_3 followed by $\text{Zn}/\text{H}_2\text{O}$ gives formaldehyde and 2-methyl propanal as products. The compound X is

- (1) 2-Methylbut-2-ene
- (2) Pent-2-ene
- (3) 3-Methylbut-1-ene
- (4) 2-Methylbut-1-ene

90. If radius of second Bohr orbit of the He^+ ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{2+} ion?

- (1) 1.587 pm
- (2) 158.7 Å
- (3) 158.7 pm
- (4) 15.87 pm

Space for Rough Work

91. In the neutral or faintly alkaline medium, KMnO_4 oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is from

- (1) +7 to +3 (2) +6 to +5
(3) +7 to +4 (4) +6 to +4

92. Given below are two statements:

Statement I: In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. $\text{HCl} + \text{ZnCl}_2$, known as Lucas Reagent.

Statement II: Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect
(2) Statement I is incorrect but Statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

93. $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$

for the above reaction at 298 K, K_c is found to be 3.0×10^{-59} . If the concentration of O_2 at equilibrium is 0.040 M then concentration of O_3 in M is

- (1) 2.4×10^{31} (2) 1.2×10^{21}
(3) 4.38×10^{-32} (4) 1.9×10^{-63}

94. Given below are two statements:

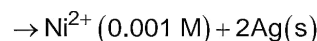
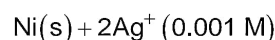
Assertion (A): In the titration of HCl against NaOH phenolphthalein is used as a suitable indicator.

Reason (R): Phenolphthalein turn to pink coloured in basic medium.

In the light of the above statements, choose the most **appropriate** answer from the options given below:

- (1) (A) is correct but (R) is not correct
(2) (A) is not correct but (R) is correct
(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

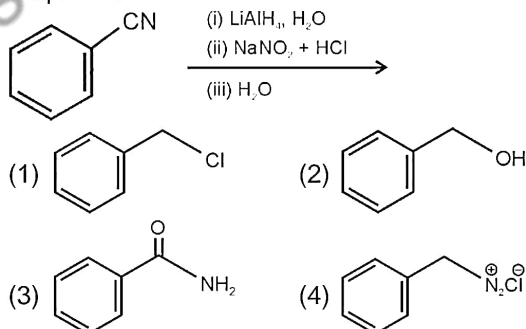
95. Find the emf of the cell in which the following reaction takes place at 298 K



$$\left(\text{Given that } E_{\text{cell}}^\circ = 1.05 \text{ V, } \frac{2.303 RT}{F} = 0.059 \text{ at } 298 \text{ K} \right)$$

- (1) 0.9615 V
(2) 1.05 V
(3) 1.0385 V
(4) 1.385 V

96. The product formed from the following reaction sequence is

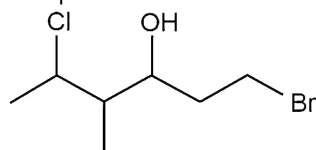


97. For a first order reaction $\text{A} \rightarrow \text{Products}$, initial concentration of A is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in min^{-1} is

- (1) 0.4606 (2) 0.2303
(3) 1.3818 (4) 0.9212

Space for Rough Work

98. The correct IUPAC name of the following compound is



- (1) 1-Bromo-4-methyl-5-chlorohexan-3-ol
 (2) 6-Bromo-4-methyl-2-chlorohexan-4-ol
 (3) 1-Bromo-5-chloro-4-methylhexan-3-ol
 (4) 6-Bromo-2-chloro-4-methylhexan-4-ol
99. The order of energy absorbed which is responsible for the color of complexes

- (A) $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$
 (B) $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$ and
 (C) $[\text{Ni}(\text{en})_3]^{2+}$

is

- (1) (C) > (A) > (B) (2) (B) > (A) > (C)
 (3) (A) > (B) > (C) (4) (C) > (B) > (A)

100. A gas is obtained by addition of dil HCl to a mixture which turns lead acetate paper black. Identify the gas.

- (1) SO_2 (2) CO_2
 (3) H_2S (4) NO_2

BOTANY

SECTION - A

101. Exoskeleton of arthropods is composed of
- (1) Chitin
 (2) Glucosamine
 (3) Cutin
 (4) Cellulose
102. Which one of the following plants does **not** show plasticity?
- (1) Buttercup
 (2) Maize
 (3) Cotton
 (4) Coriander
103. Which one of the following is **not** true regarding the release of energy during ATP synthesis through chemiosmosis? It involves
- (1) Movement of protons across the membrane to the stroma
 (2) Reduction of NADP to NADPH_2 on the stroma side of the membrane
 (3) Breakdown of proton gradient
 (4) Breakdown of electron gradient

104. Given below are two statements :

Statement I : Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance.

Statement II : Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect
 (2) Statement I is incorrect but Statement II is correct
 (3) Both Statement I and Statement II are correct
 (4) Both Statement I and Statement II are incorrect

105. Hydrocolloid carrageen is obtained from

- (1) Rhodophyceae only
 (2) Phaeophyceae only
 (3) Chlorophyceae and Phaeophyceae
 (4) Phaeophyceae and Rhodophyceae

Space for Rough Work

106. Given below are two statements:

Statement I: Decomposition is a process in which the detritus is degraded into simpler substances by microbes.

Statement II: Decomposition is faster if the detritus is rich in lignin and chitin.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

107. Which of the following organism, cell wall has stiff cellulose plates on the outer surface?

- (1) *Euglena* (2) Golden algae
- (3) Diatoms (4) *Gonyaulax*

108. Identify the **correct** set of statements :

- (a) The leaflets are modified into pointed hard thorns in *Citrus* and *Bougainvillea*
- (b) Axillary buds form slender and spirally coiled tendrils in cucumber and pumpkin
- (c) Stem is flattened and fleshy in *Opuntia* and modified to perform the function of leaves
- (d) *Rhizophora* shows vertically upward growing roots that help to get oxygen for respiration
- (e) Subaerially growing stems in grasses and strawberry help in vegetative propagation

Choose the **correct answer** from the options given below.

- (1) (b), (c), (d) and (e) Only
- (2) (a), (b), (d) and (e) Only
- (3) (b) and (c) Only
- (4) (a) and (d) Only

109. The flowers are zygomorphic in

- (a) Mustard
- (b) Gulmohar
- (c) *Cassia*
- (d) *Datura*
- (e) Chilli

Choose the **correct answer** from the options given below.

- (1) (d), (e) Only (2) (c), (d), (e) Only
- (3) (a), (b), (c) Only (4) (b), (c) Only

110. The action spectrum of photosynthesis roughly resembles the absorption spectra of

- (1) Carotenoids (2) Phycobilins
- (3) Xanthophyll (4) Chlorophyll *a*

111. Identify the **incorrect** statement related to Pollination.

- (1) Flowers produce foul odours to attract flies and beetles to get pollinated
- (2) Moths and butterflies are the most dominant pollinating agents among insects
- (3) Pollination by water is quite rare in flowering plants
- (4) Pollination by wind is more common amongst abiotic pollination

112. Given below are two statements :

Statement I : The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells.

Statement II : Mesophyll cells of C_4 plants lack RuBisCO enzyme.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

Space for Rough Work

113. Which of the following is **not** a method of *ex situ* conservation?
 (1) Micropropagation (2) Cryopreservation
 (3) *In-vitro* fertilization (4) National Parks
114. What amount of energy is released from glucose during lactic acid fermentation?
 (1) About 10% (2) Less than 7%
 (3) Approximately 15% (4) More than 18%
115. Which one of the following plants shows vexillary aestivation and diadelphous stamens?
 (1) *Allium cepa*
 (2) *Solanum nigrum*
 (3) *Colchicum autumnale*
 (4) *Pisum sativum*
116. Read the following statements and choose the set of **correct** statements
 (a) Euchromatin is loosely packed chromatin
 (b) Heterochromatin is transcriptionally active
 (c) Histone octamer is wrapped by negatively charged DNA in nucleosome
 (d) Histones are rich in lysine and arginine
 (e) A typical nucleosome contains 400 bp of DNA helix
 Choose the correct answer from the options given below.
 (1) (b), (e) Only (2) (a), (c), (e) Only
 (3) (b), (d), (e) Only (4) (a), (c), (d) Only
117. The gaseous plant growth regulator is used in plants to.
 (1) Help overcome apical dominance
 (2) Kill dicotyledonous weeds in the fields
 (3) Speed up the malting process
 (4) Promote root growth and root hair formation to increase the absorption surface
118. XO type of sex determination can be found in
 (1) Grasshoppers (2) Monkeys
 (3) *Drosophila* (4) Birds
119. Match List I with List II.

List I	List II
(a) GA ₃	(i) Gaseous PGR
(b) ABA	(ii) Terpenoids derivative
(c) Kinetin	(iii) Carotenoids derivative
(d) C ₂ H ₄	(iv) Adenine derivative

 Choose the **correct** answer from the options given below.
 (1) a-(iv), b-(i), c-(ii), d-(iii)
 (2) a-(iii), b-(i), c-(ii), d-(iv)
 (3) a-(iii), b-(iv), c-(i), d-(ii)
 (4) a-(ii), b-(iii), c-(iv), d-(i)
120. Endemism refers to
 (1) Species diversity
 (2) Species richness
 (3) Species evenness
 (4) Species confined to a region
121. Given below are two statements :
Statement I : Cleistogamous flowers are invariably autogamous.
Statement II : Cleistogamy is disadvantageous as there is no chance for cross pollination.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) Statement I is correct but Statement II is incorrect
 (2) Statement I is incorrect but Statement II is correct
 (3) Both Statement I and Statement II are correct
 (4) Both Statement I and Statement II are incorrect
122. Which of the following is **incorrectly** matched?
 (1) *Porphyra* – Floridean Starch
 (2) *Volvox* – Starch
 (3) *Ectocarpus* – Fucoxanthin
 (4) *Ulothrix* – Mannitol

Space for Rough Work

123. Which one of the following never occurs during mitotic cell division?
- Pairing of homologous chromosomes
 - Coiling and condensation of the chromatids
 - Spindle fibres attach to kinetochores of chromosomes
 - Movement of centrioles towards opposite poles
124. Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants?
- Ethylene
 - Cytokinin
 - ABA
 - Gibberellin
125. Which one of the following statements is **not true** regarding gel electrophoresis?
- The presence of chromogenic substrate gives blue coloured DNA bands on the gel.
 - Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.
 - The process of extraction of separated DNA strands from gel is called elution.
 - The separated DNA fragments are stained by using ethidium bromide.
126. The appearance of recombination nodules on homologous chromosomes during meiosis characterizes
- Sites at which crossing over occurs
 - Terminalization
 - Synaptonemal complex
 - Bivalent
127. Which of the following is **not** true for amphibolic pathway?
- When utilized as a substrate, fatty acids would be converted to acetyl CoA prior to entering the respiratory pathway.
 - In order to synthesize fatty acid, acetyl CoA needs to be removed from the respiratory pathway.
 - Once phosphoglyceric acid is formed from glycerol, it enters the respiratory pathway.
 - Before being used for respiration, carbohydrates are usually first converted into glucose.
128. The process of translation of mRNA to proteins begins as soon as
- Both the subunits join together to bind with mRNA
 - The tRNA is activated and the larger subunit of ribosome encounters mRNA
 - The small subunit of ribosome encounters mRNA
 - The larger subunit of ribosome encounters mRNA
129. What is the net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid?
- Two
 - Eight
 - Four
 - Six
130. Which one of the following statements cannot be connected to Predation?
- Both the interacting species are negatively impacted
 - It is necessitated by nature to maintain the ecological balance
 - It helps in maintaining species diversity in a community
 - Keeps prey population under control
131. Habitat loss and fragmentation, over exploitation, alien species invasion and co-extinction are causes for
- Biodiversity loss
 - Natality
 - Population explosion
 - Competition

Space for Rough Work

132. Read the following statements about the vascular bundles :

- (a) In roots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii.
- (b) Conjoint closed vascular bundles do not possess cambium
- (c) In open vascular bundles, cambium is present in between xylem and phloem
- (d) The vascular bundles of dicotyledonous stem possess endarch protoxylem
- (e) In monocotyledonous root, usually there are more than six xylem bundles present

Choose the **correct answer** from the options given below.

- (1) (a), (b), (c) and (d) Only
- (2) (a), (c), (d) and (e) Only
- (3) (a), (b), (c), (d) and (e)
- (4) (b), (c), (d) and (e) Only

133. DNA polymorphism forms the basis of

- (1) Both genetic mapping and DNA finger printing
- (2) Translation
- (3) Genetic mapping
- (4) DNA fingerprinting

134. In old trees the greater part of secondary xylem is dark brown and resistant to insect attack due to :

- (a) secretion of secondary metabolites and their deposition in the lumen of vessels.
- (b) deposition of organic compounds like tannins and resins in the central layers of stem.
- (c) deposition of suberin and aromatic substances in the outer layer of stem.
- (d) deposition of tannins, gum, resin and aromatic substances in the peripheral layers of stem.
- (e) presence of parenchyma cells, functionally active xylem elements and essential oils.

Choose the **correct** answer from the options given below.

- (1) (d) and (e) Only
- (2) (b) and (d) Only
- (3) (a) and (b) Only
- (4) (c) and (d) Only

135. **Assertion (A)** : Polymerase chain reaction is used in DNA amplification.

Reason (R) : The ampicillin resistant gene is used as a selectable marker to check transformation.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) (A) is correct but (R) is not correct
- (2) (A) is not correct but (R) is correct
- (3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

SECTION-B

136. The anatomy of springwood shows some peculiar features. Identify the **correct** set of statements about springwood.

- (a) It is also called as the earlywood
- (b) In spring season cambium produces xylem elements with narrow vessels
- (c) It is lighter in colour
- (d) The springwood along with autumn wood shows alternate concentric rings forming annual rings
- (e) It has lower density

Choose the **correct** answer from the options given below.

- (1) (a), (b) and (d) Only
- (2) (c), (d) and (e) Only
- (3) (a), (b), (d) and (e) Only
- (4) (a), (c), (d) and (e) Only

Space for Rough Work

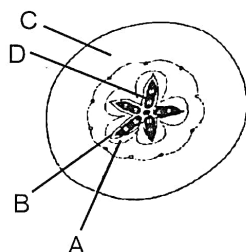
137. Match List-I with List-II.

	List-I		List-II
(a)	Metacentric chromosome	(i)	Centromere situated close to the end forming one extremely short and one very long arms
(b)	Acrocentric chromosome	(ii)	Centromere at the terminal end
(c)	Submetacentric	(iii)	Centromere in the middle forming two equal arms of chromosomes
(d)	Telocentric chromosome	(iv)	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the **correct answer** from the options given below.

- (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

138. Which part of the fruit, labelled in the given figure makes it a false fruit?



- (1) C → Thalamus
- (2) D → Seed
- (3) A → Mesocarp
- (4) B → Endocarp

139. In the following base sequences of DNA, which one can be cut easily by a particular restriction endonuclease?

- (1) 5'CTCAGT3'; 3'GAGTCA5'
- (2) 5'GTATTC3'; 3'CATAAG5'
- (3) 5'GATACT3'; 3'CTATGA5'
- (4) 5'GAATTC3'; 3'CTTAAG5'

140. What is the role of large bundle sheath cells found around the vascular bundles in C₄ plants?

- (1) To enable the plant to tolerate high temperature
- (2) To protect the vascular tissue from high light intensity
- (3) To provide the site for photorespiratory pathway
- (4) To increase the number of chloroplast for the operation of Calvin cycle

141. The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. In reference to this, which one of the following statements is false?

- (1) It allows vigorous growth of useful aerobic microbes into flocs.
- (2) Microbes consume the major part of the organic matter in the effluent
- (3) The effluent is passed into a settling tank after the BOD of sewage is increased significantly.
- (4) The sediment is pumped back into the aeration tank to serve as inoculum

Space for Rough Work

142. Match the plant with the kind of life cycle it exhibits:

	List-I		List-II
(a)	<i>Spirogyra</i>	(i)	Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
(b)	Fern	(ii)	Dominant haploid free-living gametophyte
(c)	<i>Funaria</i>	(iii)	Dominant diploid sporophyte alternating with reduced gametophyte called prothallus
(d)	<i>Cycas</i>	(iv)	Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte

Choose the **correct answer** from the options given below.

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

143. In anaerobic respiration where CO_2 is not the by product, pyruvate is reduced in presence of which reducing agent?

(1) $\text{NADPH} + \text{H}^+$ (2) FAD^+

(3) $\text{NADH} + \text{H}^+$ (4) FADH_2

144. Which one of the following will accelerate phosphorus cycle?

(1) Weathering of rocks (2) Rain fall and storms

(3) Burning of fossil fuels (4) Volcanic activity

145. If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as

(1) Expressed sequence tags

(2) Bioinformatics

(3) Sequence annotation

(4) Gene mapping

146. Which of the following occurs due to the presence of autosome linked dominant trait ?

(1) Haemophilia

(2) Thalassemia

(3) Sickle cell anaemia

(4) Myotonic dystrophy

147. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.

Reason (R) : Closely located genes assort independently.

In the light of the above statements, choose the **correct answer** from the options given below.

(1) (A) is correct but (R) is not correct

(2) (A) is not correct but (R) is correct

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

148. While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (–) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (–) for another species involved in the interaction?

(1) Commensalism (2) Competition

(3) Predation (4) Amensalism

Space for Rough Work

149. Read the following statements on lipids.

- (a) Lecithin found in the plasma membrane is a glycolipid.
- (b) Saturated fatty acids possess one or more C = C bonds.
- (c) Gingelly oil has a lower melting point, hence it remains as oil in winters.
- (d) Lipids are generally insoluble in water but soluble in some organic solvents.
- (e) When fatty acid is esterified with glycerol, monoglycerides are formed.

Choose the option with only correct statements.

- (1) (c), (d) and (e) only
 - (2) (a), (b) and (d) only
 - (3) (a), (b) and (c) only
 - (4) (a), (d) and (e) only
150. Transposons can be used during which one of the following procedures?
- (1) Autoradiography
 - (2) Gene sequencing
 - (3) Polymerase Chain Reaction
 - (4) Gene Silencing

ZOOLOGY

SECTION - A

151. *In-situ* conservation refers to

- (1) Conserve only endangered species
- (2) Conserve only extinct species
- (3) Protect and conserve the whole ecosystem
- (4) Conserve only high-risk species

152. Which of the following is **not** the function of conducting part of the human respiratory system?

- (1) Bringing temperature of the inhaled air to body temperature
- (2) Providing surface for the diffusion of O₂ and CO₂
- (3) Clearing the inhaled air from foreign particles
- (4) Humidification of inhaled air

153. Identify the asexual reproductive structure associated with *Penicillium*.

- (1) Gemmules (2) Buds
- (3) Zoospores (4) Conidia

154. Given below are two statements :

Statement I : Mycoplasma can pass through less than 1 micron filter size.

Statement II : Mycoplasma are bacteria with cell wall.

In the light of the above statements, choose the most appropriate answer from the options given below

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

155. Tegmina in cockroach arises from

- (1) Metathorax
- (2) Prothorax and mesothorax
- (3) Prothorax
- (4) Mesothorax

156. A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is C₆H₁₂O₆, then what will be the formula for maltose?

- (1) C₁₂H₂₂O₁₁ (2) C₁₂H₂₄O₁₁
- (3) C₁₂H₂₀O₁₀ (4) C₁₂H₂₄O₁₂

Space for Rough Work

157. **Assertion (A):** Osteoporosis is characterised by decreased bone mass and increased chances of fractures.
Reason (R): Common cause of osteoporosis is increased levels of estrogen.
 In the light of the above statements, choose the **most appropriate** answer from the options given below.
- (1) (A) is correct but (R) is not correct
 - (2) (A) is not correct but (R) is correct
 - (3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - (4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
158. Natural selection where more individuals acquire specific character value other than the mean character value, leads to
- (1) Disruptive change
 - (2) Random change
 - (3) Stabilising change
 - (4) Directional change
159. Given below are two statements:
Statement I: Restriction endonucleases recognise specific sequence to cut DNA known as palindromic nucleotide sequence.
Statement II: Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.
 In the light of the above statements, choose the **most appropriate** answer from the options given below.
- (1) **Statement I** is correct but **Statement II** is incorrect
 - (2) **Statement I** is incorrect but **Statement II** is correct
 - (3) Both **Statement I** and **Statement II** are correct
 - (4) Both **Statement I** and **Statement II** are incorrect
160. Which of the following is **not** a connective tissue?
- (1) Cartilage
 - (2) Neuroglia
 - (3) Blood
 - (4) Adipose tissue
161. Which of the following is a correct match for disease and its symptoms?
- (1) Myasthenia gravis – Genetic disorder resulting in weakening and paralysis of skeletal muscles
 - (2) Muscular dystrophy – An auto immune disorder causing progressive degeneration of skeletal muscles
 - (3) Arthritis – Inflamed joints
 - (4) Tetany – High Ca^{2+} level in body fluid causing rapid spasms
162. **Assertion (A) :** All vertebrates are chordates but all chordates are not vertebrates.
Reason (R) : Notochord is replaced by vertebral column in the adult vertebrates.
 In the light of the above statements, choose the **most appropriate** answer from the option given below.
- (1) (A) is correct but (R) is not correct
 - (2) (A) is not correct but (R) is correct
 - (3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - (4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
163. In gene therapy of Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because
- (1) Lymphocytes from patient's blood are grown in culture, outside the body
 - (2) Genetically engineered lymphocytes are not immortal cells
 - (3) Retroviral vector is introduced into these lymphocytes
 - (4) Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages

Space for Rough Work

164. If the length of a DNA molecule is 1.1 metres, what will be the approximate number of base pairs?
- 3.3×10^6 bp
 - 6.6×10^6 bp
 - 3.3×10^9 bp
 - 6.6×10^9 bp
165. Regarding Meiosis, which of the statements is **incorrect**?
- Pairing of homologous chromosomes and recombination occurs in Meiosis-I
 - Four haploid cells are formed at the end of Meiosis-II
 - There are two stages in Meiosis, Meiosis-I and II
 - DNA replication occurs in S phase of Meiosis-II
166. If '8' *Drosophila* in a laboratory population of '80' died during a week, the death rate in the population is _____ individuals per *Drosophila* per week.
- 1.0
 - Zero
 - 0.1
 - 10
167. In the taxonomic categories which hierarchical arrangement in descending order is **correct** in case of animals?
- Kingdom, Order, Class, Phylum, Family, Genus, Species
 - Kingdom, Order, Phylum, Class, Family, Genus, Species
 - Kingdom, Phylum, Class, Order, Family, Genus, Species
 - Kingdom, Class, Phylum, Family, Order, Genus, Species
168. Nitrogenous waste is excreted in the form of pellet or paste by
- Hippocampus*
 - Pavo*
 - Ornithorhynchus*
 - Salamandra*
169. Exploring molecular, genetic and species-level diversity for products of economic importance is called
- Bioremediation
 - Bioaccumulation
 - Bioprospecting
 - Biomagnification
170. Select the **incorrect** statement with reference to mitosis.
- Chromosomes decondense at telophase
 - Splitting of centromere occurs at anaphase
 - All the chromosomes lie at the equator at metaphase
 - Spindle fibres attach to centromere of chromosomes
171. At which stage of life the oogenesis process is initiated in human females?
- Birth
 - Adult
 - Puberty
 - Embryonic development stage
172. Detritivores breakdown detritus into smaller particles. This process is called
- Humification
 - Decomposition
 - Catabolism
 - Fragmentation
173. In an *E. coli* strain *i* gene gets mutated and its product cannot bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome?
- z*, *y*, *a* genes will not be translated
 - RNA polymerase will bind the promoter region
 - Only *z* gene will get transcribed
 - z*, *y*, *a* genes will be transcribed
174. Identify the microorganism which is responsible for the production of an immunosuppressive molecule cyclosporin A.
- Aspergillus niger*
 - Streptococcus cerevisiae*
 - Trichoderma polysporum*
 - Clostridium butylicum*

Space for Rough Work

175. Under normal physiological conditions in human beings, every 100 mL of oxygenated blood can deliver _____ mL of O_2 to the tissues.
Select the option that correctly fills the blank.
- (1) 4 mL
 - (2) 10 mL
 - (3) 2 mL
 - (4) 5 mL
176. Given below are two statements :
- Statement I :** The coagulum is formed of network of threads called thrombins.
- Statement II :** Spleen is the graveyard of erythrocytes.
- In the light of the above statements, choose the **most appropriate** answer from the options given below.
- (1) Statement I is correct but Statement II is incorrect
 - (2) Statement I is incorrect but Statement II is correct
 - (3) Both Statement I and Statement II are correct
 - (4) Both Statement I and Statement II are incorrect
177. Which of the following is present between the adjacent bones of the vertebral column in humans?
- (1) Areolar tissue
 - (2) Smooth muscle
 - (3) Intercalated discs
 - (4) Cartilage
178. Lippes loop is a type of contraceptive used as
- (1) Non-medicated IUD
 - (2) Copper releasing IUD
 - (3) Cervical barrier
 - (4) Vault barrier
179. Which of the following statements are true for spermatogenesis but do not hold true for oogenesis?
- (a) It results in the formation of haploid gametes.
 - (b) Differentiation of gamete occurs after the completion of meiosis.
 - (c) Meiosis occurs continuously in a mitotically dividing stem cell population.
 - (d) It is controlled by the Luteinising Hormone (LH) and Follicle Stimulating Hormone (FSH) secreted by the anterior pituitary.
 - (e) It is initiated at puberty
- Choose the most appropriate answer from the options given below.
- (1) (b), (d) and (e) only
 - (2) (b), (c) and (e) only
 - (3) (c) and (e) only
 - (4) (b) and (c) only
180. Select the correct pathway for transport of food in frogs.
- (1) Mouth → Buccal cavity → Pharynx → Oesophagus → Stomach → Intestine → Rectum → Cloaca
 - (2) Mouth → Buccal cavity → Oesophagus → Pharynx → Stomach → Intestine → Rectum → Cloaca
 - (3) Mouth → Buccal cavity → Pharynx → Oesophagus → Intestine → Stomach → Cloaca → Rectum
 - (4) Mouth → Buccal cavity → Oesophagus → Pharynx → Intestine → Stomach → Cloaca → Rectum
181. Choose the incorrect statement w.r.t male frogs.
- (1) Their hind limbs end in five digits.
 - (2) A copulatory pad is present on the first digit of the hind limbs.
 - (3) They capture food by a bilobed tongue.
 - (4) Their hind limbs are larger and muscular than fore limbs.

Space for Rough Work

182. Given below are two statements:

Statement I : The release of sperms into the seminiferous tubules is called spermiation.

Statement II : Spermiogenesis is the process of formation of sperms from spermatogonia.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

183. Which of the following statements with respect to endoplasmic reticulum is incorrect?

- (1) In prokaryotes only RER are present
- (2) SER are the sites for lipid synthesis
- (3) RER has ribosomes attached to ER
- (4) SER is devoid of ribosomes

184. In which of the following animals, digestive tract has additional chambers like crop and gizzard?

- (1) *Catla*, *Columba*, *Crocodilus*
- (2) *Pavo*, *Psittacula*, *Corvus*
- (3) *Corvus*, *Columba*, *Chameleon*
- (4) *Bufo*, *Balaenoptera*, *Bangarus*

185. Given below are two statements:

Statement I: Autoimmune disorder is a condition where body defense mechanism recognizes its own cells as foreign bodies.

Statement II: Rheumatoid arthritis is a condition where body does not attack self cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (1) **Statement I** is correct but **Statement II** is incorrect
- (2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

SECTION - B

186. Select the **incorrect** statement regarding synapses.

- (1) Chemical synapses use neurotransmitters.
- (2) Impulse transmission across a chemical synapse is always faster than that across an electrical synapse.
- (3) The membranes of pre-synaptic and post-synaptic neurons are in close proximity in an electrical synapse.
- (4) Electrical current can flow directly from one neuron into the other across the electrical synapse.

187. Select the **incorrect** statement with respect to acquired immunity.

- (1) Anamnestic response is due to the memory of the first encounter.
- (2) Acquired immunity is non-specific type of defense present at the time of birth.
- (3) Primary response is produced when our body encounters a pathogen for the first time.
- (4) Anamnestic response is elicited on subsequent encounters with the same pathogen.

188. Ten *E.coli* cells with ^{15}N - dsDNA are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many *E.coli* cells will have DNA totally free from ^{15}N ?

- | | |
|--------------|--------------|
| (1) 60 cells | (2) 80 cells |
| (3) 20 cells | (4) 40 cells |

189. If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness?

- | | |
|---------|----------|
| (1) 75% | (2) 100% |
| (3) 25% | (4) 50% |

Space for Rough Work

190. Which statement(s) is/are **correct** about the genetically engineered insulin?

- (a) Pro-hormone insulin contains an extra stretch of C-peptide.
- (b) A-peptide and B-peptide chains of insulin were produced separately in *E.coli*, extracted and combined by creating disulphide bonds between them.
- (c) Insulin used for treating diabetes was extracted from cattle and pigs.
- (d) Pro-hormone insulin needs to be processed for converting into a mature and functional hormone.
- (e) Some patients develop allergic reactions to the foreign insulin.

Choose the **most appropriate** answer from the options given below.

- (1) (c) and (d) only (2) (c), (d) and (e) only
- (3) (a), (b) and (d) only (4) (b) only

191. Which of the following are **not** the effects of parathyroid hormone?

- (a) Stimulates the process of bone resorption
- (b) Decreases Ca^{2+} level in blood
- (c) Reabsorption of Ca^{2+} by renal tubules
- (d) Decreases the absorption of Ca^{2+} from digested food
- (e) Increases metabolism of carbohydrates

Choose the **most appropriate** answer from the options given below.

- (1) (a) and (e) only (2) (b) and (c) only
- (3) (a) and (c) only (4) (b), (d) and (e) only

192. Match **List I** with **List II**.

	List I		List II
(a)	Bronchioles	(i)	Dense regular connective tissue
(b)	Goblet cells	(ii)	Loose connective tissue
(c)	Tendons	(iii)	Glandular tissue
(d)	Adipose Tissue	(iv)	Ciliated epithelium

Choose the **correct answer** from the options given below.

- (1) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
- (2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
- (3) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- (4) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

193. Match **List I** and **List II** with respect to methods of contraception and their respective actions.

	List I		List II
(a)	Diaphragms	(i)	Inhibit ovulation and implantation
(b)	Contraceptive pills	(ii)	Increase phagocytosis of sperm within uterus
(c)	Intra uterine devices	(iii)	Absence of menstrual cycle and ovulation following parturition
(d)	Lactational amenorrhea	(iv)	They cover the cervix blocking the entry of sperms

Choose the **correct answer** from the options given below.

- (1) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
- (2) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
- (3) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
- (4) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

194. Which of the following is **not** a desirable feature of a cloning vector?

- (1) Presence of single restriction enzyme site
- (2) Presence of two or more recognition sites
- (3) Presence of origin of replication
- (4) Presence of a marker gene

Space for Rough Work

195. Given below are two statements:

Statement I : The UTRs are present only after stop codon.

Statement II : At the end of translation, a release factor binds to the stop codon which terminates translation.

In light of the above statements, choose the **most appropriate** answer from the options given below.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and Statement II are correct
- (4) Both statement I and Statement II are incorrect

196. The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome?

- (1) a, b, c, d (2) a, c, b, d
- (3) a, d, b, c (4) d, b, a, c

197. Which of the following is a **correct** statement?

- (1) Slime moulds are saprophytic organisms classified under kingdom Monera.
- (2) Mycoplasma have DNA, ribosome and cell wall.
- (3) Cyanobacteria are a group of autotrophic organisms classified under kingdom Monera.
- (4) Bacteria are exclusively heterotrophic organisms.

198. Match **List I** with **List II**.

List I (Biological Molecules)		List II (Biological functions)	
(a)	Glycogen	(i)	Hormone
(b)	Globulin	(ii)	Biocatalyst
(c)	Steroid	(iii)	Antibody
(d)	Thrombin	(iv)	Storage product

Choose the **correct answer** from the options given below.

- (1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)
- (2) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- (3) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)
- (4) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)

199. Which one of the following statements is **correct** w.r.t. humans under normal conditions?

- (1) Blood moves freely from atrium to the ventricle during joint diastole.
- (2) Increased ventricular pressure causes closing of the semilunar valves.
- (3) The atrio-ventricular node (AVN) generates an action potential to stimulate atrial contraction.
- (4) The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria.

200. Which of the following statements is **not** true?

- (1) Homology indicates common ancestry.
- (2) Flippers of penguins and dolphins are a pair of homologous organs.
- (3) Analogous structures are a result of convergent evolution.
- (4) Sweet potato and potato is an example of analogy.

□ □ □



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